



system prompt that enforces explicit tool usage.

The prompt defines two analysis modes:

- **Verification**

When two texts are provided, the agent:

1. Executes the Verify Authorship tool.
2. Cross-checks findings using Stylistic Profile Comparison.
3. Aggregates supporting evidence from additional tools.
4. Produces a final verdict and confidence estimate.

- **Candidate-Based Attribution**

When candidate authors and a disputed text are provided, the agent:

1. Compares each candidate text against the disputed text.
2. Executes verification and stylistic comparison tools.
3. Ranks candidates according to similarity scores.
4. Reports the most likely author and confidence level.

The prompt further instructs the agent to:

- Focus on writing style rather than content.
- Report uncertainty explicitly.
- Explain conflicting evidence.
- Always invoke tools before generating conclusions.

This approach significantly reduces unsupported reasoning and improves explainability.

#### 4. Stylometric Analysis Tools

The forensic agent integrates six Python-based analysis tools.

- **Verify Authorship**

Performs overall authorship verification and produces an initial similarity estimate between two texts.

- **Vocabulary Richness Analysis**

Measures lexical diversity using indicators such as vocabulary variation and word usage patterns.

- **Stylistic Profile Comparison**

Compares linguistic fingerprints across documents and identifies stylistic similarities and differences.

- **Error Pattern Analysis**

Detects recurring spelling, punctuation, and grammatical habits that may serve as author-specific markers.

- **Frequent Word Analysis**

Identifies characteristic words and recurring lexical preferences.

- **Multi-Order N-Gram Analysis**

Extracts recurring character-level and word-level patterns that often represent stable stylistic signatures.

#### 5. Results

We used the PAN Authorship Verification dataset [3] from 2020 for the evaluation, which is widely used for authorship verification research and provides sufficiently long documents for reliable stylometric analysis.

The dataset [3] contains large number of

texts and around 400 authors, that is why it was a good choice for experimenting.

We used multiple AI models from Olama[2] and Webis for testing though the output was different for every model and there were differences in the performance where there were some models better than other models for example using minimax-m3 was more intelligent than gpt-oss and ilama 3.2 models.

Of course, we used different prompts to test our workflow and that gives different results, which are delivered in the chat in a clear and titled format so the normal user can understand it, and it contains the following points:

- Analysis type
- Summary of findings
- Supporting evidence
- Confidence score
- Limitations
- Final verdict

For an Example if we entered two texts into the chat then we ask the agent to compare the same texts if they were written by the same author then we will get the following response:

#### *Authorship Verification Report*

##### *1. Analysis Type*

*Authorship Verification — Comparing Text A (known) vs. Text B (questioned).*

##### *2. Summary*

*The two texts share a reflective, philosophical tone and first-person observation, but .....*

##### *3. Evidence*

*.....*

*Lexical sophistication — Text B's vocabulary clusters around Latinate academic terms (peculiar, manner, commonplace, exceptional, examination, distinguishes); Text A draws from everyday Anglo-Saxon vocabulary.*

##### *Convergent Features*

*Both employ reflective first-person philosophical musing.*

*Both center on perception, curiosity, and human nature.*

*Both use parallel structure ("It is a curious thing..." / "It is this restless inquiry...").*

*These convergences are genre-level (aphoristic reflection), not idiolect-level, and do not outweigh the divergences above.*

##### *4. Confidence Score: 0.72*

*Confidence is moderate-to-high that the authors differ: .....*

##### *5. Limitations*

*Sample size: ~70 words per text is below the threshold for robust statistical authorship attribution.*

*Tool unavailability: All automated stylistic tools returned errors; this analysis is based on manual linguistic inspection only.*

*....*

##### *Final Verdict*

*Different authors (or, less likely, same author in deliberately distinct registers). The contraction habit, lexical register, and sentence architecture diverge sufficiently to outweigh the surface-level thematic affinity.*

We notice that the output was much longer than the provided text, so we had to show just a part of it for demonstration.

The system performs particularly well when:

- Documents contain sufficient text length.
- Multiple tools agree on stylistic similarities.
- Distinctive lexical and character-level patterns are present.

Several limitations were observed during development:

- Some LLMs occasionally attempted to answer without tool execution.
- Python analysis required external API integration because n8n cannot directly execute local scripts.
- Model latency varied significantly across providers.
- Confidence estimates remain heuristic rather than statistically calibrated.

## 6. Conclusion

The Evaluation on the PAN 2020 Authorship Verification dataset showed that combining outputs from the six stylometric tools — covering vocabulary richness, n-gram patterns, error habits, frequent words, and stylistic profile comparison — with LLM-based synthesis produces structured, interpretable forensic reports containing explicit evidence, confidence estimates, and stated limitations, which are essential properties for any analysis

intended to support forensic or legal decision-making.

The experiments also showed that model choice materially affects both output quality and tool-adherence, with some LLMs occasionally attempting to answer without executing tools, and that confidence scores, while useful for communication, remain heuristic rather than statistically calibrated. Future work should focus on: (1) calibrating confidence estimates against ground-truth PAN labels, (2) expanding the tool suite to cover syntactic and discourse-level features, (3) enforcing tool-use guarantees at the architecture level rather than through prompting alone, and (4) testing the system across additional languages and genres to assess generalizability. Overall, these results support the broader claim that tool-augmented LLM agents offer a promising direction for more transparent and accountable AI-assisted forensic linguistics.

## 7. References

- [1] "n8n documentation." [Online]. Available: <https://docs.n8n.io>
- [2] "Ollama documentation." [Online]. Available: <https://ollama.com>
- [3] Kestemont, M., Manjavacas, E., Markov, I., Bevendorff, J., Wiegmann, M., Stamatatos, E., Potthast, M., & Stein, B. (2021). *PAN21 authorship verification: Cross-domain (test)* [Data set]. Zenodo. <https://doi.org/10.5281/zenodo.5106099>